
Feedback or Annealing?

A Controlled Study of Adaptive State Aggregation

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Abstract

Adaptive state aggregation replaces full-state Bellman updates with updates on groups of states having similar values. A natural refinement rule shrinks the group width in response to the Bellman residual, but any gain may come from annealing itself rather than from feedback. I separate the two on a 9,261-state inventory-control MDP. Because the Bellman operator contracts the residual span by γ per sweep, that span—and therefore the residual-driven aggregation width—decays geometrically. This gives a natural open-loop baseline whose decay rate is determined in advance by γ and the solver’s update schedule, rather than tuned after observing the adaptive method. Across three global/aggregate update mixes, 20 paired sampling seeds, and five compute budgets, the largest advantage feedback shows over that timetable at any budget whose bootstrap interval excludes zero is 0.2% of the control’s error, and its largest such disadvantage is 10.8%; at the final iterate every 95% interval lies inside a preregistered ± 0.02 practical-equivalence region and also contains zero. The schedule feedback induces is worth far more than the feedback: against a fixed $\varepsilon = 0.05$ it moves the error by about 23% at a matched budget, in a direction that reverses as the budget grows, so the final iterate alone reports neither effect. Full-state value iteration also reaches the ground-truth tolerance in fewer charged backups than any aggregation arm. The case study shows why adaptive dynamic programming needs fixed-resolution, matched open-loop, and decision-quality controls, at more than one budget.

1 Introduction

State aggregation combats the state-space cost of dynamic programming by solving a smaller surrogate MDP [1–3]. In the adaptive scheme of Chen et al. [4], states with nearby current values share a group, and ordinary Bellman sweeps alternate with stochastic group sweeps. A fixed bin width ε creates a familiar cost–accuracy tradeoff: coarse groups are cheap but impose approximation error, while fine groups preserve accuracy but perform more backups.

This invites a feedback controller. When the Bellman residual is large, a coarse representation may suffice; as the residual contracts, the algorithm can refine. The key identification problem is that feedback also creates an annealing schedule. Comparing it only with a fixed width cannot tell whether the residual signal helps or whether almost any schedule with the same decay would work.

I test the signal itself, against an open-loop geometric control matched at every mix in starting width, floor, and time to the floor, so that the two arms differ only in whether the width is set by the residual. The original experiment was preregistered at one update mix and a 20 ms endpoint; I preserve that result, including the endpoint’s failure, and add a work-indexed robustness check at three mixes. Three things come out of it. First, a geometric control is a consequence, not a modelling choice: span contraction makes the residual-driven width obey a geometric envelope with ratio γ per global sweep (Proposition 1), so feedback cannot produce a schedule shape an open-loop rule could not.

Second, the signal buys nothing at any of five compute budgets. Third, the schedule it induces is not innocuous: at a matched budget it moves the error by about 23% relative to never refining, in a direction that reverses as the budget grows. A final-iterate comparison reports none of this.

2 Related work

Classical hard aggregation replaces each block of states by a common value, with approximation and induced-policy losses controlled by within-block value variation [1, 3, 6]. Several methods adapt the representation during computation: Bertsekas and Castanon [2] group states by residual magnitude, Singh et al. [7] adapt soft cluster memberships, variable-resolution control splits cells on value, policy, influence, or variance criteria [8], and progressive disaggregation slices regions relative to the evolving value function [9]. I use the simpler construction of Chen et al. [4], which repeatedly bins states by current value with asymptotic error $2\varepsilon/(1 - \gamma)$.

Bellman residuals provide computable error certificates [10], their span seminorm has been optimized directly in approximate value iteration [11], and prioritized sweeping uses local errors to choose backups [12, 13]; yet a small residual need not be a good proxy for the decision objective [14], which is why policy loss is evaluated separately here. My use of the residual is narrower—its global span changes one resolution parameter, not group membership, state priority, or the fitting objective—and I propose no new abstraction or convergence theorem.

3 Method

For a finite discounted cost MDP, define

$$(TV)(s) = \min_a \left[c(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right]. \quad (1)$$

The solver bins $V(s)$ into occupied intervals of width ε . A global iteration evaluates (1) for every state. An aggregate iteration samples one state per group, performs one lifted Bellman backup for that representative, and updates the group value with step size $\alpha_t = 1/\sqrt{t}$. A schedule (ℓ_g, ℓ_a) repeats ℓ_g global and ℓ_a aggregate iterations. Rebinning occurs on entry to each aggregate phase, and lifting occurs on return to the global phase.

After each global sweep I record $r_t = \text{span}(TV_t - V_t) = \max_s (TV_t - V_t)(s) - \min_s (TV_t - V_t)(s)$. The tested feedback rule is

$$\varepsilon_t^{\text{RESIDUAL}} = \max\{\varepsilon_{\min}, cr_t\}, \quad c = 0.084, \quad \varepsilon_{\min} = 0.05. \quad (2)$$

The constant was fixed at the preregistered (2, 5) mix, to align the first residual-controlled width with the coarsest fixed baseline: there the first aggregate entry follows $\ell_g = 2$ arm-independent opening sweeps, after which $r = 5.93429$ deterministically, so $0.084r = 0.49848 \approx 0.5$. The opening sweeps are ordinary value iteration, so this width is available before the feedback arm is ever run—but it moves with ℓ_g , giving $\varepsilon_0^{\text{RESIDUAL}} = 0.400, 0.499$, and 0.538 at (5, 2), (2, 5), and (1, 20). The main control never reads r_t :

$$\varepsilon_j^{\text{GEOMETRIC}} = \max \left\{ \varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \left(\frac{\varepsilon_{\min}}{\varepsilon_0^{\text{RESIDUAL}}} \right)^{j/(C-1)} \right\}, \quad (3)$$

over aggregate entries j , with $(\varepsilon_0^{\text{RESIDUAL}}, C)$ set per mix to (0.400, 7), (0.499, 13), and (0.538, 14): the starting width, the floor, and the entry at which RESIDUAL clamps. GEOMETRIC therefore matches RESIDUAL's schedule endpoints at every mix and differs from it only in closed-loop versus open-loop refinement. A fixed $\varepsilon = 0.05$ arm tests whether refinement is useful at all; slower geometric and coarser fixed controls are retained in the artifact.

$\varepsilon_0^{\text{RESIDUAL}}$ is predictable in advance but C is not, so GEOMETRIC reads it off one RESIDUAL run: at the two added mixes GEOMETRIC is a matched control, not a preregistered one. Matching moves only the control, and moves it toward RESIDUAL, so it can only make a feedback advantage harder to claim. The preregistered control held (0.4985, 14) everywhere—the same arm at (2, 5) to within one entry; Section 5 reports what that mismatch was worth.

Table 1: Final iterate over 20 paired seeds. R/G/F are residual, matched geometric, and fixed $\varepsilon = .05$; policy loss and B(M), millions of billed backups, follow the same order. The interval is for the median of the 20 seedwise paired differences, not of the marginal medians.

Mix	$\text{err}_\infty(\text{R})$	$\text{err}_\infty(\text{G})$	$\text{err}_\infty(\text{F})$	R-G (95% CI)	Policy loss R/G/F	B(M) R/G/F
(5, 2)	.05001	.05001	.05001	$1.7\text{e}-7$ [$-2.8\text{e}-7, 9.0\text{e}-7$]	.00210/.00210/.00210	67.685/67.684/67.686
(2, 5)	.13882	.13882	.13792	$5.1\text{e}-7$ [$-9.6\text{e}-5, 2.0\text{e}-5$]	.01518/.01280/.01280	30.214/30.206/30.195
(1, 20)	.20343	.20115	.19093	$-.00201$ [$-.00320, .00541$]	.05178/.05178/.01649	9.185/9.186/9.196

Why a geometric control is the right one. GEOMETRIC’s endpoints are matched to RESIDUAL, but its *shape* is forced.

Proposition 1 (span contraction, 5, §6.6). *Along the global sweeps of any schedule, $r_{t+1} \leq \gamma r_t$. Hence $\varepsilon^{\text{RESIDUAL}} \leq \max\{\varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g}\}$, where n_g counts global sweeps since the first aggregate entry, and the width reaches its floor within $\lceil \log(\varepsilon_{\min}/\varepsilon_0^{\text{RESIDUAL}}) / \log \gamma \rceil$ of them.*

The contraction is standard [5, §6.6] and I claim no novelty for it; Appendix A reproduces the proof only to fix the constants. The consequence for ε is the point: the feedback rule inherits the contraction, so a rule reading a contracting residual is an annealing schedule whose shape is fixed before the first run and which an open-loop rule can reproduce. The aggregate phase is outside the hypothesis, so the claim is an envelope over the stretch in which the rule is still setting the width; it holds at every recorded iterate of every mix, where $\varepsilon^{\text{RESIDUAL}}$ decays by 0.698, 0.829, and 0.829 per cycle against allowed γ^{ℓ_g} of 0.774, 0.903, and 0.950. The slack grows with the aggregate share, because those phases move V too, which is why C must be measured rather than read off γ^{ℓ_g} .

Inventory MDP, protocol, and metrics. A state is the inventory vector $q \in \{-10, \dots, 10\}^3$, giving $|S| = 21^3 = 9,261$; five joint quote-aggressiveness actions trade fill probability against spread, and the one-period cost combines spread revenue with correlated inventory risk (Appendix C). I use $\gamma = .95$ and rescale costs so $\|V^*\|_\infty = 100$. This is a small market-making-style control model, not a calibrated market simulator. Each arm runs 10,000 iterations at each mix; the MDP is deterministic, so seeds 0–19 vary the within-group sampling stream only and are paired across arms. The primary metric is $\text{err}_\infty = \|V - V^*\|_\infty$, with the greedy policy π_V evaluated separately by $\|V^{\pi_V} - V^*\|_\infty$. Work-indexed curves use billed Bellman backups $B = n_g|S| + \sum_t K_t$ for K_t occupied groups, an accounting favorable to aggregation: rebinning and lifting add $(n_r + n_l)|S|$ state operations, and the residual scan is not charged. Paired differences are medians with seeded 10,000-resample percentile-bootstrap intervals, against the preregistered practical-equivalence region $[-0.02, 0.02]$ on the 100-normalized value scale.

4 Results

Once the residual and geometric rules reach $\varepsilon_{\min}$ their work–error trajectories are visually inseparable at every mix (Figure 1, Appendix B); Table 1 makes the small differences visible. At all three mixes the paired interval for residual minus matched geometric lies inside the ± 0.02 equivalence region and also contains zero, with median differences 1.7×10^{-7} , 5.1×10^{-7} , and -0.00201 : against a control matched on its own schedule endpoints, the residual signal does not move the final iterate in either direction.

Value accuracy does not order decision quality. At (2, 5) residual and matched geometric agree on value error to seven decimals yet differ by 19% in policy loss, and at (1, 20) residual trails fixed-0.05 on both ($+0.01250$ in value error, 95% CI $[0.00745, 0.01561]$; $+0.03529$ in policy loss). Feedback cannot be justified by either metric here. Median final work differs by at most 0.13% within a mix, so these comparisons are work-matched, though representation maintenance adds 39–96% for the residual arm. Full-state VI meets its 10^{-10} tolerance in 454 sweeps, or 4.20M billed backups: aggregation is not competitive here, and the experiment identifies behavior inside an approximation scheme rather than a new best solver.

Which decision was the schedule, and which was the feedback? The final iterate says whether feedback moves the endpoint, not whether refinement scheduling does anything at all: by 10,000 iterations every scheduled arm has sat at $\varepsilon_{\min}$ for almost the whole run, and at (5, 2) the residual

Table 2: Median paired difference in value error at fixed compute budgets, as a percentage of the control’s error there, over 20 seeds; negative favors residual. RESIDUAL–FIXED isolates refinement scheduling, RESIDUAL–GEOMETRIC the feedback signal given the schedule. * marks a cell whose 95% bootstrap interval excludes zero. Construction: Appendix D.

Mix	Contrast	10^5	3×10^5	10^6	3×10^6	final
(5, 2)	RESIDUAL–FIXED	+0.0	+0.2*	−0.1	−0.0	+0.0
(5, 2)	RESIDUAL–GEOMETRIC	+0.0	+0.0	−0.2*	+0.0	+0.0
(2, 5)	RESIDUAL–FIXED	+0.1	−23.5*	−14.1*	+0.4*	+0.6*
(2, 5)	RESIDUAL–GEOMETRIC	−0.1	−0.1	+0.2*	−0.0	+0.0
(1, 20)	RESIDUAL–FIXED	−18.8*	−5.1*	−5.4*	+10.2*	+6.5*
(1, 20)	RESIDUAL–GEOMETRIC	+4.5*	+10.8*	+3.4*	−1.4	−1.0

and fixed arms end 3.8×10^{-8} apart. Re-reading the same runs at four intermediate budgets, where the schedules are still live, separates the two decisions (Table 2; differences are relative to the control, since the ± 0.02 margin says nothing where both arms sit at error 10). Feedback’s largest resolved advantage over the matched timetable at any budget is 0.2% of the control’s error and its largest resolved disadvantage 10.8%, while refinement scheduling is worth about 23%—two orders of magnitude more than the advantage—and reverses: at (1, 20) annealing is 18.8% better than never refining at 10^5 backups and 10.2% worse at 3×10^6 . The schedule does real work in both directions while the feedback generating it contributes nothing, as Proposition 1 predicts. The large RESIDUAL–FIXED entries fall in the steep descent, where a fixed budget lands between iterates; their sign and order of magnitude are robust, their exact percentages are not.

The preregistered endpoint failed. The original primary endpoint, (2, 5) at 20 ms, gave a residual-minus-geometric median of -0.0417 with 95% CI $[-0.0459, 0.0011]$, indeterminate under the equivalence rule, and repeated executions changed the interval because 20 ms lies on a steep transient and selects a different iterate as machine load changes. I report rather than replace that outcome; every backup-indexed analysis here is a robustness analysis motivated by the failure, not a retroactively preregistered endpoint.

5 What the negative result isolates

Endpoint matching is load-bearing. An open-loop control must be matched on the clock on which adaptation occurs, here aggregate entries rather than total iterations: transplanted unchanged to (1, 20), the slow geometric arm never finishes its timetable and ends ten times worse than every other arm, and recalibrating only its cycle count restores it (Appendix E).

An unmatched control manufactures an effect. Held at (0.4985, 14) at every mix, the preregistered open-loop arm differs from residual at (5, 2) by .00632, 95% CI $[\text{.006319}, \text{.006327}]$: narrow enough to exclude zero, with all 20 seeds favoring the control. Read alone that is a real and precisely measured difference; it is also entirely an artifact of the control, which opens 25% coarser than the residual rule there and takes twice as long to clamp. Re-anchoring on those two numbers moves the same comparison to 1.7×10^{-7} with an interval containing zero. The trap in negative results with low simulation variance is not that real differences go undetected, but that abundant precision resolves differences far too small to justify the mechanism and charges them to whichever knob was left unmatched.

6 Limitations and conclusion

One deterministic instance, one residual statistic, one floor, and one feedback constant support no claim about other MDPs; the mixes, budget ladder, and backup endpoints postdate the preregistered experiment, and billed backups omit representation maintenance. Feedback has room to matter past Proposition 1’s envelope: heterogeneous regional residuals, an unknown floor, or a horizon ending before the schedule does. Within this scope, residual feedback never improves on an open-loop timetable matched to the schedule it induces, because such a rule already is one—worth about 23% at a matched budget, with a budget-dependent sign. Attributing that gain needs controls reproducing the induced path, at more than one endpoint; otherwise ordinary scheduling is mistaken for intelligence.

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A Proof of Proposition 1

The span-seminorm contraction below is standard [5, §6.6] and is reproduced only to fix the constants the envelope uses.

Fix V, U and a state s , and let a^* attain the minimum in $(TU)(s)$. Evaluating the bracket of (1) at a^* rather than at the minimizer for V gives

$$\begin{aligned} (TV)(s) &\leq c(s, a^*) + \gamma \sum_{s'} P(s' | s, a^*) V(s') = (TU)(s) + \gamma \sum_{s'} P(s' | s, a^*) (V - U)(s') \\ &\leq (TU)(s) + \gamma \max_{s'} (V - U)(s'), \end{aligned}$$

and the symmetric argument, taking a^* to attain the minimum in $(TV)(s)$, gives $(TV)(s) \geq (TU)(s) + \gamma \min_{s'} (V - U)(s')$. Both bounds hold at every s , so $(TV - TU)(s)$ lies in $[\gamma \min(V - U), \gamma \max(V - U)]$ and therefore

$$\text{span}(TV - TU) \leq \gamma \text{span}(V - U).$$

Along a global phase $V_{t+1} = TV_t$, so $r_{t+1} = \text{span}(TV_{t+1} - V_{t+1}) = \text{span}(TV_{t+1} - TV_t) \leq \gamma \text{span}(V_{t+1} - V_t) = \gamma \text{span}(TV_t - V_t) = \gamma r_t$. Iterating over the n_g global sweeps taken since the first aggregate entry and substituting into (2) gives the stated envelope, and the floor is reached once $\varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g} \leq \varepsilon_{\min}$.

The aggregate phase is outside the hypothesis: it replaces V by a value lifted from group representatives, which is not an application of T and can raise the span. The envelope is therefore a claim about the stretch in which the rule is still setting the width, and it is verified empirically over exactly that stretch, at every recorded iterate of all three mixes.

B Work-indexed trajectories

Rules reaching $\varepsilon_{\min}$ have similar error floors — inventory $N = 3, Q = 10, \gamma = 0.95$

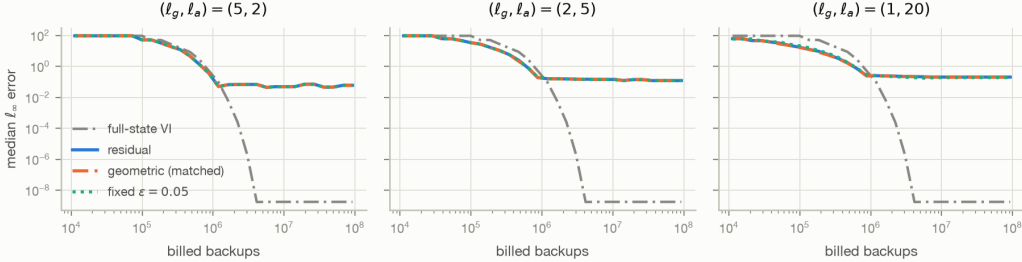


Figure 1: Median value error against billed Bellman backups; terminal estimates are retained past an arm’s 10,000-iteration horizon. Residual, fast geometric, and fixed-0.05 nearly coincide within each mix. The deterministic VI trajectory is repeated for reference; its convergence check stops after 454 sweeps (4.20M backups) though the plotted diagnostic runs to the common horizon.

The flat right-hand tails are held terminal values, not continued computation: each arm stops at its own horizon (67.7M, 30.2M, and 9.2M billed backups at (5, 2), (2, 5), and (1, 20)), and the plotted value is carried forward so the three panels share an axis. Comparisons at budgets past an arm’s own horizon are therefore not work comparisons, which is why the ladder in Appendix D stops at 3×10^6 .

C Inventory instance

The five joint quote-aggressiveness actions have fill probabilities (.1, .2, .4, .6, .8) and spreads (1, .8, .5, .3, .15), so aggressive quotes fill often at a worse price. The one-period cost is

$$c(q, a) = 0.02 q^\top \Sigma q - \delta_a f_a L(q),$$

where δ_a and f_a are the spread and fill probability of action a , Σ has unit diagonal and correlation 0.5 so inventory risk is correlated across the three assets, and $L(q)$ removes fills that would leave the box $\{-10, \dots, 10\}^3$. Costs are rescaled after solving so that $\|V^*\|_\infty = 100$, which puts ε , the equivalence margin, and every reported error on one interpretable scale.

D Budget-indexed differences: construction

Each cell of Table 2 is the median over 20 seeds of the per-seed difference, divided by the control’s error for that seed at the same budget; forming the ratio per seed before aggregating preserves the pairing. Errors are read as a step function of billed backups: the error the budget actually delivered, not the one the run was on its way to. The ladder stops at 3×10^6 because the cheapest mix, $(1, 20)$, bills only 9.2M backups in total, and rungs past an arm’s own horizon would compare held terminal values rather than work. Absolute differences and seeded 10,000-resample percentile-bootstrap intervals for both the absolute and relative medians are in the artifact.

E The transplanted slow schedule

The preregistered slow geometric arm used 1,429 aggregate cycles, exactly the number induced by 10,000 iterations at cycle length seven. Transplanted unchanged to $(1, 20)$, whose cycle length is 21, it receives only 477 aggregate entries and is still decaying at the horizon: it ends at $\varepsilon = .232$ and error 1.59, an order of magnitude worse than every other arm. Recalibrating only its cycle count to 477, changing nothing else, ends it at $\varepsilon = .05024$ and error .209, back in the neighborhood of residual and fast geometric. The apparent failure was an unfinished timetable, not missing feedback, and it is the reason the recalibrated arm rather than the frozen one is carried as the slow control at mixes other than the preregistered $(2, 5)$.